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Research Interests

Climate sensitivity, cloud feedback, radiative forcing

Education

Ph.D. Atmospheric Sciences
University of Washington, 2010
M.S. Atmospheric Sciences
University of Washington, 2007
B.S. Meteorology
Pennsylvania State University, 2004

Professional Experience

Research Scientist

Lawrence Livermore National Laboratory
2013–present

Postdoctoral Research Scholar

Lawrence Livermore National Laboratory
2011–2013

Graduate Research Assistant

University of Washington, Dept. of Atmospheric Sciences
2004–2010

Submitted Publications

Asterisks indicate current or former advisees

- Beydoun, H., N. Mahfouz, A. Varble, C. R. Terai, L.-W. Chao*, **M. D. Zelinka**, L. Bertagna, N. D. Keen, P. M. Caldwell, 2026: Thermodynamic Buffering of the Tropical Cloud Fraction Response to Warming in a Global Storm-Resolving Model, *Geophys. Res. Lett.*, submitted.
- Norris, J, S. Rahimi, L. Huang, C. Thackeray, B. Bass, **M. D. Zelinka**, and A. Hall, 2026: High vulnerability of Western United States mountain ranges to heavy rainfall events under warming, *Commun. Earth Environ.*, submitted.

Peer-Reviewed Publications

Asterisks indicate current or former advisees

114. Myers, T. A. *, T. Auerenon, D. T. McCoy, **M. D. Zelinka**, J. M. Nugent, H. Brown, 2026: Cloud-Controlling Factors

Reveal Mean-State Controls and Structural Biases in Earth System Model Stratocumulus Physics, *Geophys. Res. Lett.*, in press.

113. Thackeray, C., J. Norris, A. Hall, **M. D. Zelinka**, and S. Po-Chedley, 2026: Record setting heavy precipitation occurrence in 2024, *Sci. Adv.*, in press.
112. Ceppi, P., Bodas-Salcedo, A., **Zelinka, M. D.**, Andrews, T., Brient, F., Chadwick, R., Dai, A.-Z., Gregory, J. M., Hwang, Y.-T., Kang, S. M., Kay, J. E., Mauritsen, T., Ogura, T., Tselioudis, G., Watanabe, M., Webb, M. J., and Wing, A. A., 2026: The Cloud Feedback Model Intercomparison Project (CFMIP) contribution to CMIP7, *Geosci. Model Dev.*, 19, 8693–8708, doi:10.5194/gmd-19-8693-2026.
111. Van Loon, S., M. Rugenstein, **M. D. Zelinka**, and T. Andrews, 2026: Recent Weakening of the Global Radiative Feedback, *Geophys. Res. Lett.*, 53, doi:10.1029/2026GL122980.
110. Dingley, B., et al. including **M. D. Zelinka**, 2026: CMIP7 Data Request: Atmosphere Priorities and Opportunities, *Geosci. Model Dev.*, 19, 2945–2984, doi:10.5194/gmd-19-2945-2026.
109. **Zelinka, M. D.**, T. A. Myers*, Y. Qin*, L.-W. Chao*, S. A. Klein, S. Po-Chedley, P.-L. Ma, C. Wall, P. Ceppi, A. Gettelman, 2026: Recent cloud trends and extremes reaffirm established bounds on cloud feedback and aerosol-cloud interactions, *Commun. Earth Environ.*, 7, 615, doi:10.1038/s43247-026-03461-8.
108. Tam, R. Y. S., T. A. Myers*, **M. D. Zelinka**, C. Proistosescu, Y.-J. Lin, K. Marvel, 2026: Meteorological Drivers of the Low-Cloud Radiative Feedback Pattern Effect and its Uncertainty, *Atmos. Chem. Phys.*, 26, 4289–4311, doi:10.5194/acp-26-4289-2026.
107. Chao, L.-W.*, **M. D. Zelinka**, C. R. Terai, H. Beydoun, B. R. Hillman, N. D. Keen, P. M. Caldwell, P. A. Bogenschutz, and S. A. Klein, 2026: Examining Cloud Feedback Components in the Simple Cloud-Resolving E3SM Atmosphere Model (SCREAM), *J. Climate*, 39, 2563–2578, doi: 10.1175/JCLI-D-25-0656.1.
106. Zhang, Y., et al. including **M. D. Zelinka**, 2026: Understanding Biases in Simulated Cloud Radiative Effects in E3SMv3, *Earth Space Sci.*, doi:10.1029/2025EA004757.
105. Feng, C., et al. including **M. D. Zelinka**, 2026: Interconnection of Aerosol Cloud Interactions and Cloud Feedback through Warm Rain Process, *J. Geophys. Res.*, 131, doi:10.1029/2025JD044854.
104. Ceppi, P., S. Wilson Kemsley, H. Andersen, T. Andrews, R. J. Kramer, P. Nowack, C. J. Wall, and **M. D. Zelinka**, 2026: Emerging low-cloud feedback and adjustment in global satellite observations, *Atmos. Chem. Phys.*, 26, 4153–4171, doi:10.5194/acp-26-4153-2026.
103. Espinosa, Z.*, **M. D. Zelinka**, C. M. Bitz, K. C. Armour, 2026: The Strength of Coupling to the Southern Ocean Modulates Tropical Eastern Pacific Variability and Forced Response, *J. Climate*, 39, 1543–1558, doi:10.1175/JCLI-D-25-0226.1.
102. Nugent, J. M., et al. including **M. D. Zelinka**, 2026: Overview of the Nephela Perturbed Parameter Ensemble for aerosol-cloud interactions in E3SMv3, *J. Adv. Model. Earth Syst.*, 18, doi:10.1029/2025MS004989.
101. Fridlind, A. M., et al. including **M. D. Zelinka**, 2025: Toward a Climate OSSE Framework for Satellite Mission Design, *Bull. Amer. Meteor. Soc.*, 107, E431–E445, doi:10.1175/BAMS-D-24-0242.1.
100. Aeronson, T. et al. including **M. D. Zelinka**, 2025: Global models predict clouds at the wrong time of day: does it matter for radiation and climate? *Sci. Adv.*, 11, eady3236, doi:10.1126/sciadv.ady3236.
99. Zehrung, A., A. D. King, Z. Nicholls, **M. D. Zelinka**, and M. Meinshausen, 2025: Standardising the “Gregory method” for calculating equilibrium climate sensitivity, *Geosci. Model Dev.*, 18, 9433–9450, doi:10.5194/gmd-18-9433-2025.
98. Chao, L.-W.*, **M. D. Zelinka**, et al., 2025: Evaluating Mean State Cloud Properties in the Simple Cloud-Resolving E3SM Atmosphere Model (SCREAM), *J. Adv. Model. Earth Syst.*, 17, doi:10.1029/2025MS005133.
97. Hill, P. G., D. L. Finney, and **M. D. Zelinka**, 2025: Cloud feedback uncertainty in the equatorial Pacific across CMIP6 models, *Geophys. Res. Lett.*, 52, doi:10.1029/2025GL117183.
96. Terai, C. R., et al. including **M. D. Zelinka**, 2025: Radiative, hydrologic, and circulation responses to warming in Cess-Potter simulations using the global 3.25-km SCREAM, *J. Adv. Model. Earth Syst.*, 17, doi:10.1029/2024MS004930.
95. Qin, Y.*, P.-L. Ma, **M. D. Zelinka**, S. A. Klein, T. Zhang, X. Zheng, V. E. Larson, M. Huang, 2025: Impact of Turbulence Representation on the Relationship Between Cloud Feedback and Aerosol-Cloud Interaction in an E3SMv2 Perturbed Parameter Ensemble, *J. Adv. Model. Earth Syst.*, 17, doi:10.1029/2024MS004756.
94. Mauritsen, T., et al. including **M. D. Zelinka**, 2025: Earth’s energy imbalance more than doubled in recent decades, *AGU adv.*, 6, doi:10.1029/2024AV001636.
93. Zhou, C.*, Q. Wang, I. Tan, L. Zhang, **M. D. Zelinka**, M. Wang, and J. Bloch-Johnson, 2025: Sea ice pattern effect

- on Earth's energy budget is characterized by hemispheric asymmetry, *Sci. Adv.*, 11, doi:10.1126/sciadv.adr4248.
92. **Zelinka, M. D.**, L.-W. Chao*, T. A. Myers*, Y. Qin*, and S. A. Klein, 2025: Technical note: Recommendations for diagnosing cloud feedbacks and rapid cloud adjustments using cloud radiative kernels, *Atmos. Chem. Phys.*, 25, 1477-1495, doi:10.5194/acp-25-1477-2025.
 91. Bonan, D. B., J. E. Kay, N. Feldl, and **M. D. Zelinka**, 2025: Mid-latitude clouds contribute to Arctic amplification via interactions with other climate feedbacks, *Environ. Res.: Climate*, 4, 015001, doi:10.1088/2752-5295/ada84b.
 90. Lin, Y.-J., G. V. Cesana, C. Proistosescu, **M. D. Zelinka**, and K. C. Armour, 2025: The relative importance of forced and unforced temperature patterns in driving the time variation of low-cloud feedback, *J. Climate*, 38, 513-529, doi:10.1175/JCLI-D-24-0014.1.
 89. Thackeray, C., **M. D. Zelinka**, J. Norris, A. Hall, S. Po-Chedley, 2024: Relationship between tropical cloud feedback and climatological bias in clouds, *Geophys. Res. Lett.*, 51, doi:10.1029/2024GL111347.
 88. Ceppi, P., T. A. Myers*, P. Nowack, C. J. Wall, and **M. D. Zelinka**, 2024: Implications of a pervasive climate model bias for low-cloud feedback, *Geophys. Res. Lett.*, 51, doi:10.1029/2024GL110525.
 87. Espinosa, Z.* and **M. D. Zelinka**, 2024: The Shortwave Cloud-SST Feedback Amplifies Multi-Decadal Pacific Sea Surface Temperature Trends: Implications for Observed Cooling, *Geophys. Res. Lett.*, 51, doi:10.1029/2024GL111039.
 86. Lee, J. et al. including **M. D. Zelinka**, 2024: Systematic and Objective Evaluation of Earth System Models: PCMDI Metrics Package (PMP) version 3, *Geosci. Model Dev.*, 17, 3919-3948, doi:10.5194/gmd-17-3919-2024.
 85. Tan, I., **M. D. Zelinka**, Q. Coopman, B. H. Kahn, L. Oreopoulos, G. Tselioudis, D. T. McCoy, N. Li, 2024: Contributions from cloud morphological changes to the interannual shortwave cloud feedback based on MODIS and ISCCP satellite observations, *J. Geophys. Res.*, 129, doi:10.1029/2023JD040540.
 84. Cesana, G. V., et al. including **M. D. Zelinka**, 2024: Observational constraint on a feedback from supercooled clouds reduces projected warming uncertainty, *Commun. Earth Environ.*, 5, 181, doi:10.1038/s43247-024-01339-1.
 83. Zhao, X. et al including **M. D. Zelinka**, 2024: Larger cloud liquid water enhances both aerosol indirect forcing and cloud radiative feedback in two Earth System Models, *Geophys. Res. Lett.*, 51, doi:10.1029/2023GL105529.
 82. Chao, L.-W.*, **M. D. Zelinka**, and A. E. Dessler, 2024: Evaluating Cloud Feedback Components in Observations and Their Representation in Climate Models, *J. Geophys. Res.*, 129, doi:10.1029/2023JD039427.
 81. Qin, Y.*, X. Zheng, S. A. Klein, **M. D. Zelinka**, P.-L. Ma, J.-C. Golaz, S. Xie, 2024: Causes of Reduced Climate Sensitivity in E3SM from Version 1 to Version 2, *J. Adv. Model. Earth Syst.*, 16, doi:10.1029/2023MS003875.
 80. Rugenstein, M., **M. D. Zelinka**, K. Karnauskas, P. Ceppi, and T. Andrews, 2023: Patterns of Surface Warming Matter for Climate Sensitivity, *Eos*, 104, doi:10.1029/2023EO230411.
 79. Samset, B., C. Zhou*, J. S. Fuglestad, M. T. Lund, J. Marotzke, and **M. D. Zelinka**, 2023: Steady global surface warming from 1973 to 2022 but increased warming rate after 1990, *Commun. Earth Environ.*, 4, 400, doi:10.1038/s43247-023-01061-4.
 78. Bonan, D. B., N. Feldl, **M. D. Zelinka**, and L. C. Hahn, 2023: Contributions to regional precipitation change and its polar-amplified pattern under warming, *Environ. Res.: Climate*, 2, 035010, doi:10.1088/2752-5295/ace27a.
 77. **Zelinka, M. D.**, C. J. Smith, Y. Qin*, and K. E. Taylor, 2023: Comparison of methods to estimate aerosol effective radiative forcings in climate models, *Atmos. Chem. Phys.*, 23, 8879-8898, doi:10.5194/acp-23-8879-2023.
 76. Myers, T. A.*, **M. D. Zelinka**, and S. A. Klein, 2023: Observational Constraints on the Cloud Feedback Pattern Effect, *J. Climate*, 36, 6533-6545, doi:10.1175/JCLI-D-22-0862.1.
 75. Zhou, C.*, M. Wang, **M. D. Zelinka**, Y. Liu, Y. Dong, K. C. Armour, 2023: Explaining Forcing Efficacy with Pattern Effect and State Dependence, *Geophys. Res. Lett.*, 50, doi:10.1029/2022GL101700.
 74. **Zelinka, M. D.**, I. Tan, L. Oreopoulos, G. Tselioudis, 2023: Detailing Cloud Property Feedbacks with a Regime-Based Decomposition, *Clim Dyn.*, 60, 2983-3003, doi:10.1007/s00382-022-06488-7.
 73. Santer, B. D., et al. including **M. D. Zelinka**, 2022: Robust anthropogenic signal identified in the seasonal cycle of tropospheric temperature, *J. Climate*, 35(18), 6075-6100, doi:10.1175/JCLI-D-21-0766.1.
 72. Hausfather, Z., K. Marvel, G. A. Schmidt, J. W. Nielsen-Gammon, and **M. D. Zelinka**, 2022: Climate simulations: recognize the "hot model" problem, *Nature*, doi:10.1038/d41586-022-01192-2.
 71. Samset, B., C. Zhou*, J. Fuglestad, M. Lund, J. Marotzke, **M. D. Zelinka**, 2022: Earlier emergence of a temperature response to mitigation by filtering annual variability, *Nat Commun.*, 13, 1578, doi:10.1038/s41467-022-29247-y.

70. McCoy, D. M. et al. including **M. D. Zelinka**, 2022: Extratropical shortwave cloud feedbacks in the context of the global circulation and hydrological cycle, *Geophys. Res. Lett.*, 49, doi:10.1029/2021GL097154.
69. Ma, P.-L., et al. including **M. D. Zelinka**, 2022: Better calibration of cloud parameterizations and subgrid effects increases the fidelity of E3SM Atmosphere Model version 1, *Geosci. Model Dev.*, 15, 2881–2916, doi:10.5194/gmd-15-2881-2022.
68. Qin, Y.*, **M. D. Zelinka**, and S. A. Klein, 2022: On the Correspondence between Atmosphere-Only and Coupled Simulations for Radiative Feedbacks and Forcing from CO₂, *J. Geophys. Res.*, 127, doi:10.1029/2021JD035460.
67. **Zelinka, M. D.**, S. A. Klein, Y. Qin*, and T. A. Myers*, 2022: Evaluating climate models' cloud feedbacks against expert judgment, *J. Geophys. Res.*, 127, doi:10.1029/2021JD035198.
66. Hahn L. C., K. C. Armour, **M. D. Zelinka**, C. M. Bitz, and A. Donohoe, 2021: Contributions to Polar Amplification in CMIP5 and CMIP6 Models. *Front. Earth Sci.*, doi: 10.3389/feart.2021.710036.
65. Muelmenstaedt, J., M. Salzmann, J. E. Kay, **M. D. Zelinka**, P. L. Ma, S. Hornig, and J. Quaas, 2021: An underestimated negative cloud feedback from cloud lifetime changes, *Nature Clim. Change*, 11, 508–513, doi:10.1038/s41558-021-01038-1.
64. Santer, B. D., et al. including **M. D. Zelinka**, 2021: Using climate model simulations to constrain observations, *J. Climate*, doi:10.1175/JCLI-D-20-0768.1.
63. Myers, T. A.*, R. C. Scott, **M. D. Zelinka**, S. A. Klein, J. R. Norris, and P. M. Caldwell, 2021: Observational Constraints on Low Cloud Feedback Reduce Uncertainty of Climate Sensitivity, *Nature Clim. Change*, doi:10.1038/s41558-021-01039-0.
62. Thackeray, C. W., A. Hall, **M. D. Zelinka**, and C. G. Fletcher, 2021: Assessing prior emergent constraints on surface albedo feedback in CMIP6, *J. Climate*, 34(10), 3889–3905, doi:10.1175/JCLI-D-20-0703.1.
61. Po-Chedley, S., B. D. Santer, S. Fueglistaler, **M. D. Zelinka**, P. J. Cameron-Smith, J. F. Painter, and Q. Fu, 2021: Natural variability can explain model-satellite differences in tropical tropospheric warming, *Proc. Natl. Acad. Sci.*, doi:10.1073/pnas.2020962118.
60. Pihl, E., et al. including **M. D. Zelinka**, 2021: 10 New Insights in Climate Science 2021 – a Horizon Scan, *Global Sustainability*, 1–65, doi:10.1017/sus.2021.2.
59. Zhou, C.*, **M. D. Zelinka**, A. E. Dessler, and M. Wang, 2021: Greater committed warming after accounting for the SST pattern effect, *Nature Clim. Change*, 11, 132–136, doi:10.1038/s41558-020-00955-x.
58. Ma, H.-Y., et al. including **M. D. Zelinka**, 2021: A multi-year short-range hindcast experiment with CESM1 for evaluating climate model moist processes from diurnal to interannual timescales, *Geosci. Model Dev.*, 14, 73–90, doi:10.5194/gmd-14-73-2021.
57. McCoy, D. M., P. Field, A. Bodas-Salcedo, G. S. Elsaesser, and **M. D. Zelinka**, 2020: A regime-oriented approach to observationally constraining extratropical shortwave cloud feedbacks, *J. Climate*, doi:10.1175/JCLI-D-19-0987.1.
56. Sherwood, S., et al. including **M. D. Zelinka**, 2020: An Assessment of Earth's Climate Sensitivity Using Multiple Lines of Evidence, *Rev. Geophys.*, 58, doi:10.1029/2019RG000678.
55. Scott, R. C., T. A. Myers*, J. R. Norris, **M. D. Zelinka**, S. A. Klein, M. Sun, and D. R. Doelling, 2020: Observed Sensitivity of Low-Cloud Radiative Effects to Meteorological Perturbations over the Global Oceans. *J. Climate*, 33, 7717–7734, doi:10.1175/JCLI-D-19-1028.1.
54. Dong, Y. K. C. Armour, **M. D. Zelinka**, C. Proistosescu, D. S. Battisti, C. Zhou*, and T. Andrews, 2020: Intermodel Spread in the Pattern Effect and Its Contribution to Climate Sensitivity in CMIP5 and CMIP6 Models. *J. Climate*, 33, 7755–7775, doi:10.1175/JCLI-D-19-1011.1.
53. **Zelinka, M. D.**, T. A. Myers*, D. T. McCoy, S. Po-Chedley, P. M. Caldwell, P. Ceppi, S. A. Klein, and K. E. Taylor, 2020: Causes of higher climate sensitivity in CMIP6 models, *Geophys. Res. Lett.*, 47, doi:10.1029/2019GL085782.
52. Zhou, C.*, Y. Hu, J. Lu, and **M. D. Zelinka**, 2020: Responses of the Hadley Circulation to regional sea surface temperature changes, *J. Climate*, 33, 429–441, doi:10.1175/JCLI-D-19-0315.1.
51. Po-Chedley, S., **M. D. Zelinka**, N. Jeevanjee, T. J. Thorsen, and B. D. Santer, 2019: Climatology explains inter-model spread in upper tropospheric cloud and relative humidity response to greenhouse warming, *Geophys. Res. Lett.*, 46, doi:10.1029/2019GL084786.
50. Santer, B. D., et al. including **M. D. Zelinka**, 2019: Quantifying stochastic uncertainty in detection time of human-caused climate signals, *Proc. Natl. Acad. Sci.*, 116 (40) 19821–19827, doi:10.1073/pnas.1904586116.
49. Chen, Y.-J., Y.-T. Hwang, **M. D. Zelinka**, and C. Zhou*, 2019: Distinct patterns of cloud changes associated with decadal variability and their contribution to observed cloud cover trends, *J. Climate*, 32, 7281–7301,

doi:10.1175/JCLI-D-18-0443.1.

48. Zhang, Y., et al. including **M. D. Zelinka**, 2019: Evaluation of Clouds in Version 1 of the E3SM Atmosphere Model with Satellite Simulators, *J. Adv. Model. Earth Syst.*, 11, 1253–1268, doi:10.1029/2018MS001562.
47. Golaz, J.-C., et al. including **M. D. Zelinka**, 2019: The DOE E3SM coupled model version 1: Overview and evaluation at standard resolution, *J. Adv. Model. Earth Syst.*, 11, doi:10.1029/2018MS001603.
46. Santer, B. D., et al. including **M. D. Zelinka**, 2019: Celebrating the anniversary of three key events in climate change science, *Nature Clim. Change*, 9, 180–182, doi:10.1038/s41558-019-0424-x.
45. Terai, C. R., Y. Zhang, S. A. Klein, **M. D. Zelinka**, J. C. Chiu, and Q. Min, 2019: Mechanisms behind the extratropical stratiform low-cloud optical depth response to temperature in ARM site observations, *J. Geophys. Res.*, 124, doi:10.1029/2018JD029359.
44. McCoy, D. T., et al. including **M. D. Zelinka**, 2019: Cloud feedbacks in extratropical cyclones: insight from long-term satellite data and high-resolution global simulations, *Atmos. Chem. Phys.*, 19, 1147–1172, doi:10.5194/acp-19-1147-2019.
43. Colman, R., J. R. Brown, C. Franklin, L. Hanson, H. Ye, and **M. D. Zelinka**, 2019: Evaluating cloud feedbacks and rapid responses in the ACCESS model, *J. Geophys. Res.*, 124, doi:10.1029/2018JD029189.
42. **Zelinka, M. D.**, K. M. Grise, S. A. Klein, C. Zhou*, A. M. DeAngelis, and M. W. Christensen, 2018: Drivers of the Low Cloud Response to Poleward Jet Shifts in the North Pacific in Observations and Models, *J. Climate*, 31, 7925–7947, doi:10.1175/JCLI-D-18-0114.1.
41. Santer, B. D., et al. including **M. D. Zelinka**, 2018: Human influence on the seasonal cycle of tropospheric temperature, *Science*, 361, doi:10.1126/science.aas8806.
40. Caldwell, P. M., **M. D. Zelinka**, and S. A. Klein, 2018: Evaluating Emergent Constraints on Equilibrium Climate Sensitivity, *J. Climate*, 31, 3921–3942, doi:10.1175/JCLI-D-17-0631.1.
39. Po-Chedley, S., et al. including **M. D. Zelinka**, 2018: Sources of intermodel spread in the lapse rate and water vapor feedbacks, *J. Climate*, 31, 3187–3206, doi:10.1175/JCLI-D-17-0674.1.
38. Qu, X., A. Hall, A. M. DeAngelis, **M. D. Zelinka**, S. A. Klein, H. Su, B. Tian, and C. Zhai, 2018: On proposed emergent constraints of climate sensitivity, *J. Climate*, 31, 863–875, doi:10.1175/JCLI-D-17-0482.1.
37. Tsushima, Y., et al. including **M. D. Zelinka**, 2017: The Cloud Feedback Model Intercomparison Project (CFMIP) Diagnostic Codes Catalogue – metrics, diagnostics and methodologies to evaluate, understand and improve the representation of clouds and cloud feedbacks in climate models, *Geosci. Model Dev.*, 10, 4285–4305, doi:10.5194/gmd-10-4285-2017.
36. **Zelinka M. D.**, D. A. Randall, M. J. Webb, and S. A. Klein, 2017: Clearing clouds of uncertainty, *Nature Clim. Change* 7, 674–678 doi:10.1038/nclimate3402.
35. Zhou, C.*, **M. D. Zelinka**, and S. A. Klein, 2017: Analyzing the dependence of global cloud feedback on the spatial pattern of sea surface temperature change with a Green's Function approach, *J. Adv. Model. Earth Syst.*, 9, 2174–2189, doi:10.1002/2017MS001096.
34. C. Bonfils, et al. including **M. D. Zelinka**, 2017: Competing influences of anthropogenic warming, ENSO, and plant physiology on future terrestrial aridity, *J. Climate*, 30, 6883–6904, doi:10.1175/JCLI-D-17-0005.1.
33. Ceppi, P., F. Brient, **M. D. Zelinka**, and D. L. Hartmann, 2017: Cloud feedback mechanisms and their representation in global climate models, *WIREs Climate Change*, e465, doi:10.1002/wcc.465.
32. Zhou, C.*, **M. D. Zelinka**, and S. A. Klein, 2016: Impact of decadal cloud variations on the Earth's energy budget, *Nature Geoscience*, 9, 871–874, doi:10.1038/ngeo2828.
31. **Zelinka, M. D.**, C. Zhou*, and S. A. Klein, 2016: Insights from a Refined Decomposition of Cloud Feedbacks, *Geophys. Res. Lett.*, 43, 9259–9269, doi:10.1002/2016GL069917.
30. Terai, C., S. A. Klein, and **M. D. Zelinka**, 2016: Constraining the low-cloud optical depth feedback at middle and high latitudes using satellite observations, *J. Geophys. Res.*, 121, 9696–9716, doi:10.1002/2016JD025233.
29. Norris, J. R., R. J. Allen, A. T. Evan, **M. D. Zelinka**, C. W. O'Dell, and S. A. Klein, 2016: Evidence for Climate Change in the Satellite Cloud Record, *Nature*, 536, 72–75, doi:10.1038/nature18273.
28. McCoy, D. T., I. Tan, D. L. Hartmann, **M. D. Zelinka**, T. Storelvmo, 2016: On the relationships among cloud cover, mixed-phase partitioning, and planetary albedo in GCMs, *J. Adv. Model. Earth Syst.*, 8, 650–668, doi:10.1002/2015MS000589.
27. Tan, I., T. Storelvmo, and **M. D. Zelinka**, 2016: Observational constraints on mixed-phase clouds imply higher climate sensitivity, *Science*, 352, 6282, 224–227, doi:10.1126/science.aad5300.
26. Yuan, T., L. Oreopoulos, **M. D. Zelinka**, H. Yu, J. Norris, M. Chin, S. Platnick, and K. Meyer, 2016: Positive low

- cloud and dust feedbacks amplify tropical North Atlantic multidecadal oscillation, *Geophys. Res. Lett.*, 43, 1349–1356, doi:10.1002/2016GL067679.
25. Caldwell, P. M., **M. D. Zelinka**, K. E. Taylor, and K. Marvel, 2016: Quantifying the Sources of Inter-Model Spread in Equilibrium Climate Sensitivity, *J. Climate*, 29, 513–524, doi:10.1175/JCLI-D-15-0352.1.
 24. Santer, B. D., S. Solomon, D. Ridley, J. Fyfe, F. Beltran, C. Bonfils, J. Painter, and **M. D. Zelinka**, 2016: Volcanic effects on climate, *Nature Clim. Change*, 6, 3–4, doi:10.1038/nclimate2859.
 23. Zhou, C. *, **M. D. Zelinka**, A. E. Dessler, S. A. Klein, 2015, The relationship between inter-annual and long-term cloud feedbacks, *Geophys. Res. Lett.*, 42, 10,463–10,469, doi:10.1002/2015GL066698.
 22. DeAngelis, A. M., X. Qu, **M. D. Zelinka**, and A. Hall, 2015: An observational radiative constraint on hydrologic cycle intensification, *Nature*, 528, 249–253, doi:10.1038/nature15770.
 21. McCoy, D. T., et al. including **M. D. Zelinka**, 2015: Mixed-phase cloud physics and Southern Ocean cloud feedback in climate models, *J. Geophys. Res.*, 120, 9539–9554, doi: 10.1002/2015JD023603.
 20. Marvel, K. et al. including **M. D. Zelinka**, 2014: External influences on modeled and observed cloud trends, *J. Climate*, 28, 4820–4840, doi:10.1175/JCLI-D-14-00734.1.
 19. Santer, B. D., et al. including **M. D. Zelinka**, 2015: Observed multi-variable signals of late 20th and early 21st century volcanic activity, *Geophys. Res. Lett.*, 42, 500–509, doi:10.1002/2014GL062366.
 18. Zhou, C. *, A. E. Dessler, **M. D. Zelinka**, P. Yang, and T. Wang, 2014: Cirrus feedback on inter-annual climate fluctuations, *Geophys. Res. Lett.*, 41, doi:10.1002/2014GL062095.
 17. Johnston, M. S., et al. including **M. D. Zelinka**, 2014: Diagnosing the average spatio-temporal impact of convective systems - Part 2: A model intercomparison using satellite data, *Atmos. Chem. Phys.*, 14, 8701–8721, doi:10.5194/acp-14-8701-2014.
 16. **Zelinka, M. D.**, T. Andrews, P. M. Forster, and K. E. Taylor, 2014: Quantifying Components of Aerosol-Cloud-Radiation Interactions in Climate Models, *J. Geophys. Res.*, 119, 7599–7615, doi:10.1002/2014JD021710.
 15. Ceppi, P., **M. D. Zelinka**, and D. L. Hartmann, 2014: The Response of the Southern Hemispheric Eddy-Driven Jet to Future Changes in Shortwave Radiation in CMIP5, *Geophys. Res. Lett.*, 41, 3244–3250, doi:10.1002/2014GL060043.
 14. Caldwell, P. M., et al. including **M. D. Zelinka**, 2014: Statistical Significance of Climate Sensitivity Predictors Obtained by Data Mining, *Geophys. Res. Lett.*, 41, 1803–1808, doi:10.1002/2014GL059205.
 13. Santer, B. D., et al. including **M. D. Zelinka**, 2014: Volcanic Contribution to Decadal Changes in Tropospheric Temperature, *Nature Geoscience*, doi:10.1038/ngeo2098.
 12. Johnston, M. S., et al. including **M. D. Zelinka**, 2013: Diagnosing the average spatio-temporal impact of convective systems - Part 1: A methodology for evaluating climate models, *Atmos. Chem. Phys.*, 13, 12043–12058, doi:10.5194/acp-13-12043-2013.
 11. Grise, K.M., L.M. Polvani, G. Tselioudis, Y. Wu, and **M.D. Zelinka**, 2013: The ozone hole indirect effect: Cloud-radiative anomalies accompanying the poleward shift of the eddy-driven jet in the Southern Hemisphere. *Geophys. Res. Lett.*, 40, 1–5, doi:10.1002/grl.50675.
 10. **Zelinka, M.D.**, S.A. Klein, K.E. Taylor, T. Andrews, M.J. Webb, J.M. Gregory, and P.M. Forster, 2013: Contributions of Different Cloud Types to Feedbacks and Rapid Adjustments in CMIP5. *J. Climate*. 26, 5007–5027. doi: 10.1175/JCLI-D-12-00555.1.
 9. Zhou, C. *, **M.D. Zelinka**, A.E. Dessler, P. Yang, 2013: An analysis of the short-term cloud feedback using MODIS data. *J. Climate*. 26, 4803–4815. doi:10.1175/JCLI-D-12-00547.1.
 8. Klein, S.A., et al. including **M. D. Zelinka**, 2013: Are climate model simulations of clouds improving? An evaluation using the ISCCP simulator. *J. Geophys. Res.* 118, 1329–1342. doi:10.1002/jgrd.50141.
 7. Forster, P.M., et al. including **M. D. Zelinka**, 2013: Evaluating adjusted forcing and model spread for historical and future scenarios in the CMIP5 generation of climate models. *J. Geophys. Res.* 118, 1139–1150. doi: 10.1002/jgrd.50174.
 6. **Zelinka, M.D.**, S.A. Klein, and D.L. Hartmann, 2012: Computing and Partitioning Cloud Feedbacks Using Cloud Property Histograms. Part I: Cloud Radiative Kernels. *J. Climate*, 25, 3715–3735. doi:10.1175/JCLI-D-11-00248.1.
 5. **Zelinka, M.D.**, S.A. Klein, and D.L. Hartmann, 2012: Computing and Partitioning Cloud Feedbacks Using Cloud Property Histograms. Part II: Attribution to Changes in Cloud Amount, Altitude, and Optical Depth. *J. Climate*, 25, 3736–3754. doi:10.1175/JCLI-D-11-00249.1.

4. **Zelinka, M.D.** and D.L. Hartmann, 2012: Climate Feedbacks and their Implications for Poleward Energy Flux Changes in a Warming Climate. *J. Climate*, 25, 608–624, doi:10.1175/JCLI-D-11-00096.1.
3. **Zelinka, M.D.** and D.L. Hartmann, 2011: The Observed Sensitivity of High Clouds to Mean Surface Temperature Anomalies in the Tropics. *J. Geophys. Res.*, 116, D23103, doi:10.1029/2011JD016459.
2. **Zelinka, M.D.** and D.L. Hartmann, 2010: Why is Longwave Cloud Feedback Positive? *J. Geophys. Res.*, 115, D16117, doi:10.1029/2010JD013817.
1. **Zelinka, M.D.** and D.L. Hartmann, 2009: Response of Humidity and Clouds to Tropical Deep Convection. *J. Climate*, 22, 2389–2404. doi:10.1175/2008JCLI2452.1.

Book Chapters

6. Chao, L. -W.*, A. E. Dessler, and **M. D. Zelinka**, 2026: Climate Feedbacks. In: Robinson, W.A. (Ed.), *Encyclopedia of Atmospheric Sciences*, vol. 5. Elsevier, Academic Press, pp. 117–128, doi:10.1016/B978-0-323-96026-7.00038-2.
5. McCoy, D. T., M. E. Frazer, J. Muelmenstaedt, I. Tan, C. R. Terai, and **M. D. Zelinka**, 2024: Extratropical Cloud Feedbacks, in *Clouds and their Climatic Impacts: Radiation, Circulation, and Precipitation*, S. C. Sullivan (Ed) and C. Hoose (Ed), American Geophysical Union.
4. Contributing author to Forster et al. 2021: The Earth's Energy Budget, Climate Feedbacks, and Climate Sensitivity, in *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*. Masson-Delmotte et al. (eds.). Cambridge University Press, Cambridge, United Kingdom and New York, NY, USA, pp. 923-1054, doi:10.1017/9781009157896.009.
3. McCoy, D.T., D. L. Hartmann, and **M. D. Zelinka**, 2017: Mixed-Phase Cloud Feedbacks, in *Mixed-phase Clouds: Observations and Modeling*, Andronache, C. (Ed.), Elsevier.
2. Tan, I., T. Storelvmo, and **M. D. Zelinka**, 2017: The climatic impact of thermodynamic phase partitioning in mixed-phase clouds, in *Mixed-phase Clouds: Observations and Modeling*, Andronache, C. (Ed.), Elsevier.
1. Dessler, A.E. and **M. D. Zelinka**, 2015: Climate Feedbacks, in *Encyclopedia of Atmospheric Sciences, 2nd edition*, Vol 2, pp. 18-25, G. R. North (editor-in-chief), J. Pyle and F. Zhang (editors).

Recent Honors & Awards

- Richard P. and Linda S. Turco Lectureship, Department of Climate, Meteorology, & Atmospheric Sciences, University of Illinois Urbana–Champaign, 2024-2025.
- LLNL Physical and Life Sciences Directorate Awards for Excellence in Publications for Terai et al. (2025), Lee et al. (2024), Sherwood et al. (2020), Zhou et al. (2016), and Norris et al. (2016)
- Zelinka et al. (2020) named "Editor's Choice Paper" by Editor-in-Chief of *Geophysical Research Letters*, 2024
- American Meteorological Society Henry G. Houghton Award, 2022
- Eos Research Spotlights for Zelinka et al. (2022), Zelinka et al. (2020), Zhou et al. (2017), Zelinka et al. (2016), and McCoy et al. (2016)
- Editors' Citation for Excellence in Refereeing - *Geophysical Research Letters*, 2021
- LLNL Deputy Director's Science & Technology Excellence in Publication Awards for Zelinka et al. (2020), Sherwood et al. (2020), and Santer et al. (2015)
- Sherwood et al. (2020) named runner-up for *Science Magazine's* 2020 Breakthrough of the Year
- *Nature Climate Change* Research Highlight for Dong et al. (2020)
- US CLIVAR Research Highlight for Zelinka et al. (2020)
- LLNL Outstanding Mentor Award, 2018
- LLNL Early and Mid-Career Recognition Program Award, 2018
- *Diablo Magazine* 40 Under 40, 2017
- LLNL Spot Award for "Excellence in Publications", 2017

Professional Service and Leadership

- Co-chair, 2027 Gordon Research Conference on Radiation and Climate

- Vice Chair, 2025 Gordon Research Conference on Radiation and Climate
- Editor, *Journal of Climate*, Dec 2024–present
- Steering Group, CMIP7 Data Request (Atmosphere Theme), Jul 2024–present
- External Peer Reviewer, NASA Langley Research Center Science Directorate, Nov 2023
- Programme Advisory Group, Uncertainty in Climate Sensitivity due to Clouds (CloudSense) Research Programme, Sep 2023–present
- CFMIP Scientific Steering Committee, Jul 2023–present
- Discussion Leader, 2023 Gordon Research Conference on Radiation & Climate
- AGU Global Environmental Change Fellows Committee, 2022–2024
- Contributing Author, IPCC 6th Assessment Report, 2019–2021
- LLNL Physical and Life Sciences Postdoc Committee, 2017–2020
- Section Editor, *Current Climate Change Reports* Topical Collection on Climate Feedbacks, 2014–2017
- Contributor, climatefeedback.org, 2017–present
- Chair, 2011 Gordon Research Seminar on Radiation and Climate
- Proposal reviewer for DOE, European Research Council, NASA, and NSF
- Editor, *Proceedings of the National Academy of Sciences*
- Reviewer for:
Atmosphere | *Atmos. Ocean* | *Atmos. Chem. Phys.* | *Atmos. Meas. Tech.* | *Atmos. Sci. Lett.*
B Am Meteorol Soc | *Clim. Dynam.* | *Climatic Change* | *Earth's Future* | *Earth Syst. Dyn.*
Earth Syst. Sci. Data | *Environ. Res. Lett.* | *Geophys. Res. Lett.* | *Geosci. Model Dev.*
Int. J. Appl. Earth Obs. Geoinf. | *J. Adv. Model. Earth Syst.* | *J. Appl. Meteorol. Clim.*
J. Atmos. Oceanic Technol. | *J. Atmos. Sci.* | *J. Climate* | *J. Geophys. Res.* | *J. Meteorol. Soc. Jpn.*
Nature | *Nat. Clim. Change* | *Nat. Commun.* | *Nat. Geosci.* | *npj Climate and Atmospheric Science*
P. Natl. Acad. Sci. | *Sci. Rep.* | *Sci. Adv.* | *Surv. Geophys.*
- Reviewer for 29 journals including flagship journals (*Nature*, *Science*, *P. Natl. Acad. Sci.*) and leading atmospheric/climate science journals (*J. Climate*, *Geophys. Res. Lett.*, *J. Geophys. Res.*, *Atmos. Chem. Phys.*, *J. Adv. Model. Earth Syst.*, and others)

Recent Invited Presentations

- ESA-JAXA EarthCARE Workshop (Keynote), 8 Jun 2026
- GEWEX Earth's Energy Imbalance Assessment Workshop, 4 Jun 2026
- Libera Science Team Meeting, 19 May 2026
- UC San Diego, Scripps Institution of Oceanography, 27 Jan 2026
- McGill University, Department of Atmospheric and Oceanic Sciences, 21 Oct 2025
- Duke University, Earth & Climate Sciences Division, 3 Oct 2025
- Stanford University, Department of Earth System Science, 2 Apr 2025
- Univ. of Illinois Urbana–Champaign, Dept. of Climate, Meteorology, & Atmospheric Sciences, 22 Oct 2024
- CERES Science Team Meeting, 2 Oct 2024
- LLNL Senior Staff Meeting, 1 Jul 2024
- NASA Ames Research Center, Earth Science Division, 20 Jun 2024
- Yale University, School of the Environment, 25 Apr 2024
- AGU Fall Meeting: The Flows of Energy Through the Climate System Session, 12 Dec 2023
- AGU Fall Meeting: Climate Sensitivity and Feedbacks Session, 11 Dec 2023
- University of Cambridge Centre for Atmospheric Science, Department of Chemistry, 24 Oct 2023
- Yale University School of the Environment, 27 Apr 2023
- AGU Fall Meeting: Atmospheric Physics, Radiation, Clouds, and Aerosols Session, 15 Dec 2022

Advising

Postdoctoral Researchers

- Li-Wei Chao (Postdoc, Tufts University)
- Yi Qin (Research Scientist, Laoshan Laboratory)
- Timothy Myers (Research Scientist, University of Wyoming)
- Chen Zhou (Associate Professor, Nanjing University)

Graduate Students

- Zac Espinosa (Google DeepMind)

Undergraduate Students

- Russell Hunter (Duke University / Second Lieutenant, US Space Force)
- Thea Moellerstedt (Graduate Student, UCSD)
- Scott Feldman (Meteorologist, Verisk Weather Solutions)

Outreach

- LLNL Ambassador Lecturer, Aug 2024 - present
- LLNL Technical Lightning Talk: Community Cohorts program for new hires, 24 Aug 2023.
- LLNL - Las Positas College Science and Engineering Seminar Series, 20 Apr 2023.
- Science on Saturday, The Future in Focus: Predicting Climate Change through Observations, Modeling, and Artificial Intelligence, 4 Mar 2023.
- Castro Valley Rotary Club, 13 Sep 2022.
- Science on Saturday, The Future in Focus: Predicting Climate Change through Observations, Modeling, and Artificial Intelligence, 26 Feb 2022.
- San Joaquin County Office of Education Climate Change Summit, 26 Sep 2020
- Panel Discussant, Wild and Scenic Film Festival, Bankhead Theater, Livermore, CA, Jan 2020
- Univ. of Washington Dept. of Atmospheric Sciences and Program on Climate Change Outreach Teams, Sep 2004 – Dec 2010

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